GH3036 HR Performance Test Report

Goodix Confidential For lierda_hd@lierda.com Only

Revision: 1.0

Release Date: 2025-11-14

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Preface

Purpose

GH3036 HR performance test report, with the purpose of reflecting GH3036 HR performance test results.

Audience

This document is intended for:

- Software developers
- Systems and product engineers
- Technical writers

Release Notes

This document is the frist release for product version Package_V00.02.00.00.

Revision History

Revision	Date	Description
1.0	2025-11-14	First release. Heart Rate Algorithm Version EXC_2.0.3.0.

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1 Config Introduction

Software version: Package_V00.02.00.00, mainly includes:

1. APP version: GHealth_1.0.9.13.apk
2. Algorithm version:

   GH_HR_exc_pv_v2.0.3.0
3. Firmware version: GR551x_VX.X_DEMO_DRV_LIB_GH3036.bin
4. Driver library version: GH(M)3X_DEMO_v0.2.1.0_ALGO.bin

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2 Test Coverage

Table 2-1 Test Coverage Introduction

Coverage Dimension	Condition
Environment	Office indoor, Outdoor, Stadium
Temperature	27℃ indoor; 25-32℃ outside
Gender of Tester	Male & Female
Age of Tester	20~40
BMI of Tester	18~30kg/㎡
SkinTone Level of Tester	Level 2~5
Reference Device	Polar H7/H10

Note:

Skin Tone Guide:

The Pantone SkinTone Guide										
Goodix Index	1	2	3	4	5	6	7	8	9	10
Pantone Index	1Y01SP	5Y01SP	2Y06SP	4Y07SP	2Y10SP	3Y11SP	2Y13SP	3R13SP	1R14SP	4R15SP

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3 Test Result

1. Sample Number: Same as recorded in the table；

Test Method: Wear GH3036 Demo properly (with moderate tightness to ensure no slipping during exercise) with the wearing position at least 1 finger width up from the wrist bone. Then perform tests in different scenarios.

2. Test Scene:

   1) Resting: Keep relatively still and collect data;
   2) Outdoor walking: Walk outdoors and collect data on the "walk-rest" process after rest;
   3) Treadmill: Collect data on the "walk-run-walk-run-rest" process by adjusting the treadmill speed;
   4) Outdoor jogging: Collect data on the "jogging-variable speed running-jogging-resting" process outdoors;
   5) Outdoor cycling: Collect data on the "relaxed cycling-fast cycling – relaxed cycling-cycling with high resistance level- relaxed cycling " process;
   6) Wrist skipping: Jump continuously with an ordinary skipping rope. Keep jumping for 1 minute and then take a reset for 30 seconds. Repeat the test again and collect heart rate data during the process;

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3.1 Yellow People Performance Result without Confidence Check

Table 3-1 Yellow People Performance Result without Confidence Check

Test Scene	Data Amount	Algo Version	Output Time(s)	≤±3BPM	≤±5BPM	≤±10BPM	≤±15BPM	MAE
Treadmill	24	Exclusive	9.96	93.73%	97.74%	99.75%	99.95%	1.14
Outdoor jogging	37	Exclusive	11.54	91.08%	94.55%	97.42%	98.44%	1.58
Outdoor walking	10	Exclusive	11.8	88.88%	94.37%	98.49%	99.65%	1.65
Outdoor cycling	12	Exclusive	10.5	80.41%	87.12%	94.48%	98.06%	2.46
Wrist skipping	32	Exclusive	10.22	66.07%	76.04%	88.10%	94.25%	4.11

3.2 Black People Performance Result without Confidence Check

Table 3-2 White People Performance Result without Confidence Check

Test Scene	Data Amount	Algo Version	Output Time(s)	≤±3BPM	≤±5BPM	≤±10BPM	≤±15BPM	MAE
Resting	10	Exclusive	9.60	96.99%	98.61%	99.63%	99.93%	0.43
Treadmill	16	Exclusive	11.44	82.74%	89.80%	93.75%	95.05%	2.35
Outdoor jogging	16	Exclusive	12.38	81.86%	86.37%	91.49%	94.06%	6.42
Outdoor cycling	15	Exclusive	14.57	70.25%	79.32%	88.16%	91.30%	3.81
Wrist skipping	14	Exclusive	12.08	50.19%	59.42%	71.26%	78.55%	11.13

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4 Test Scene Legend

4.1 Resting

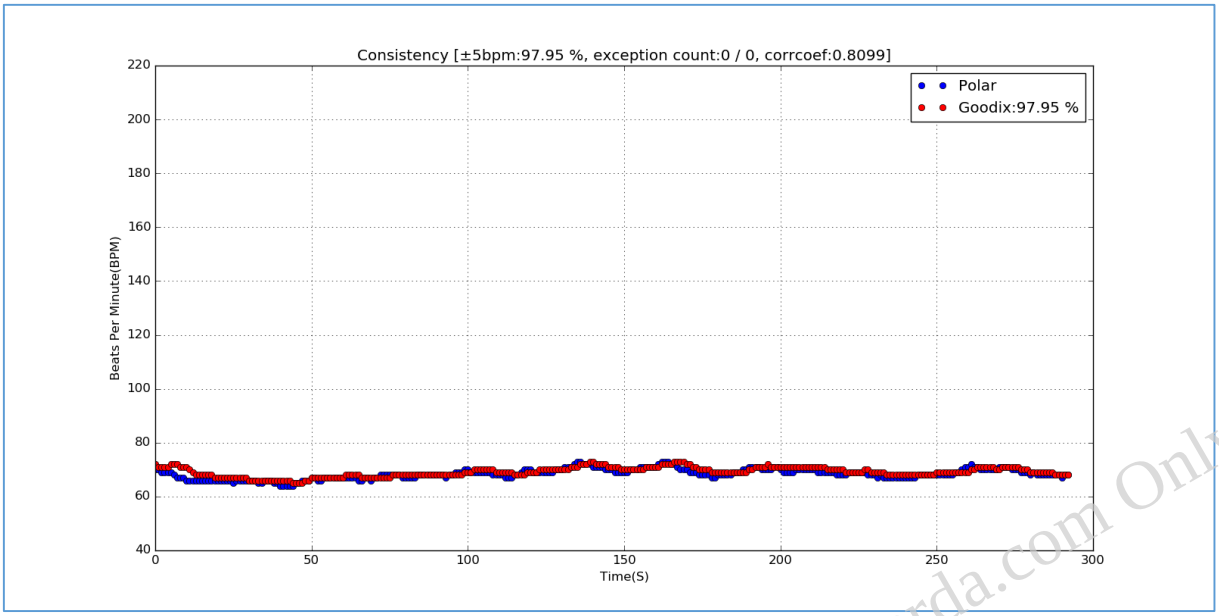

Figure 4-1 Resting

4.2 Outdoor walking

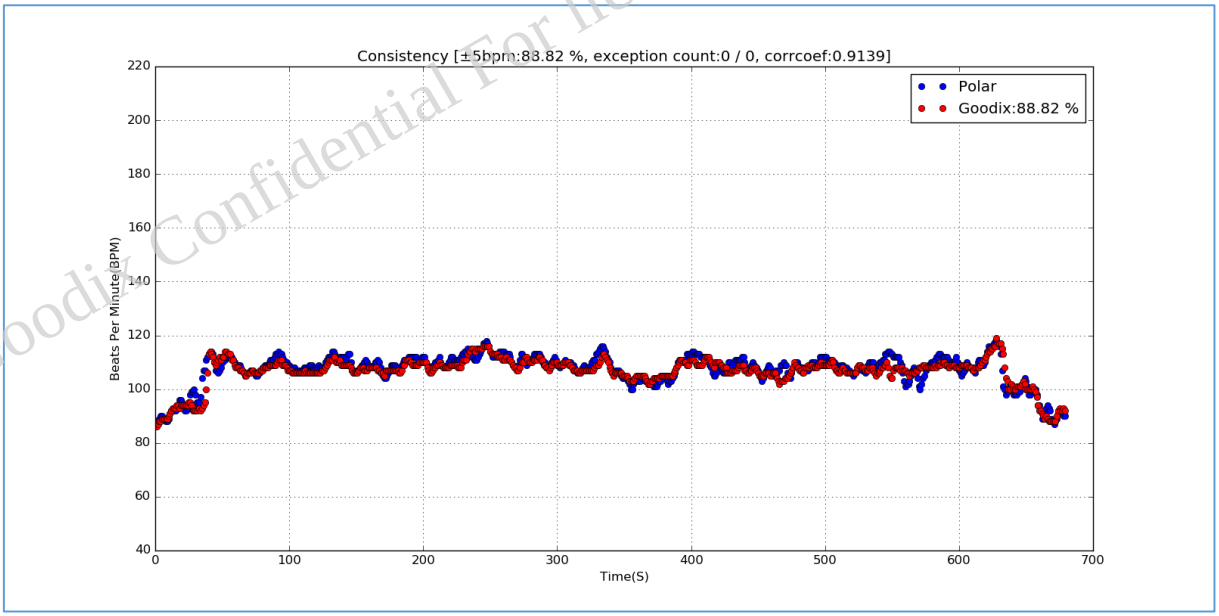

Figure 4-2 Outdoor walking

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4.3 Treadmill

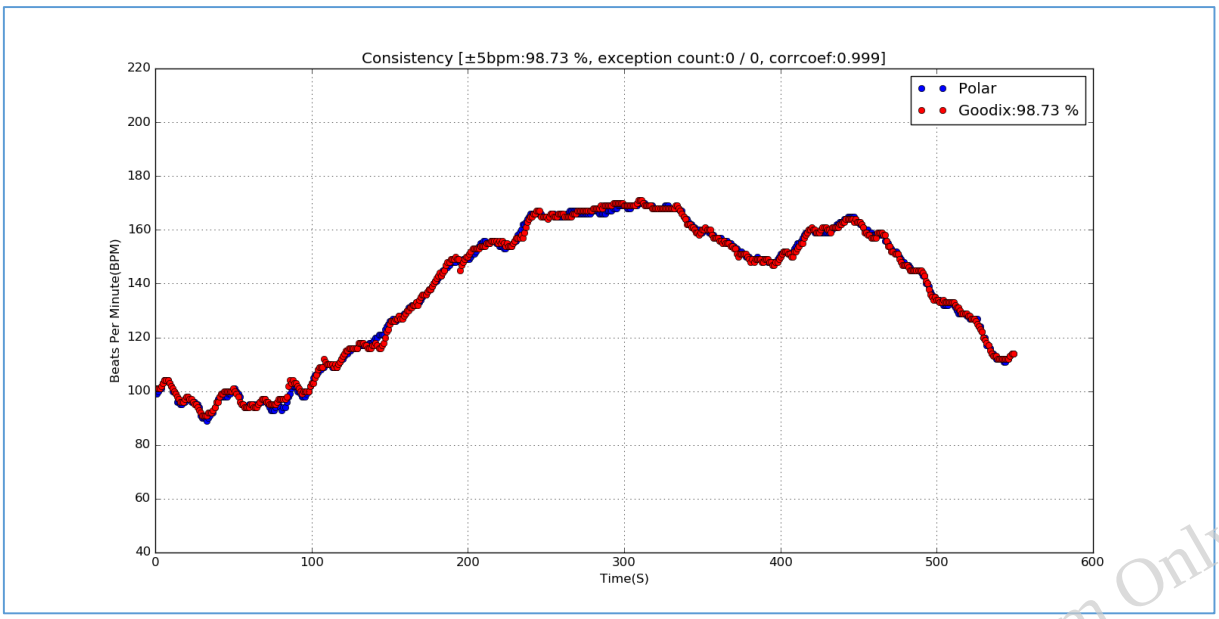

Figure 4-3 Treadmill

4.4 Outdoor jogging

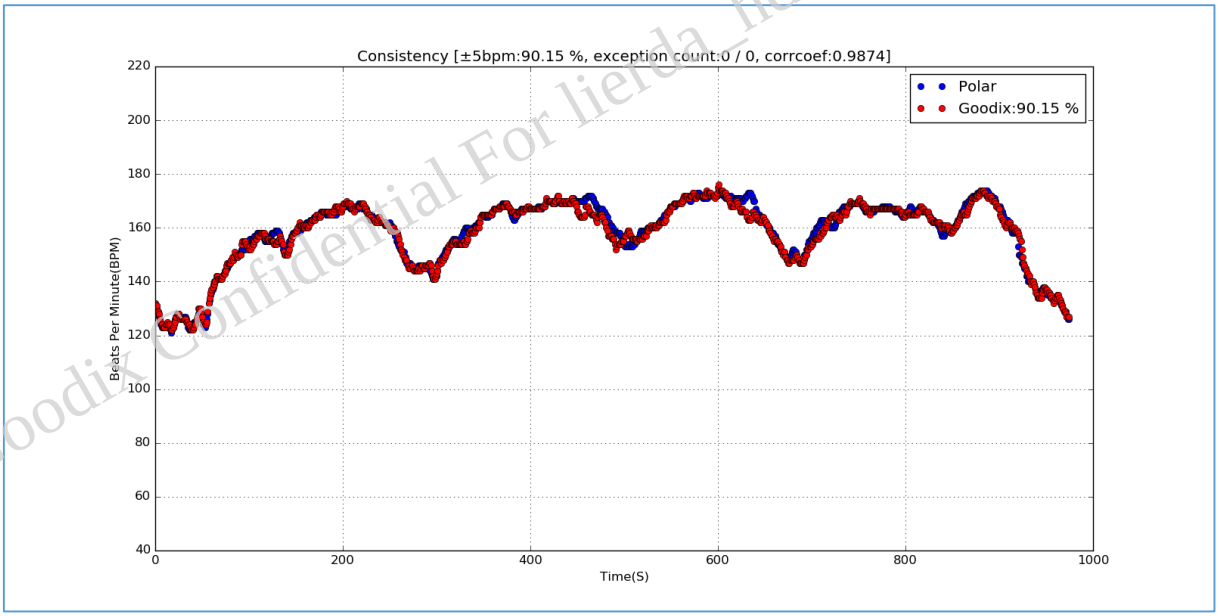

Figure 4-4 Outdoor jogging

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4.5 Outdoor bicycle

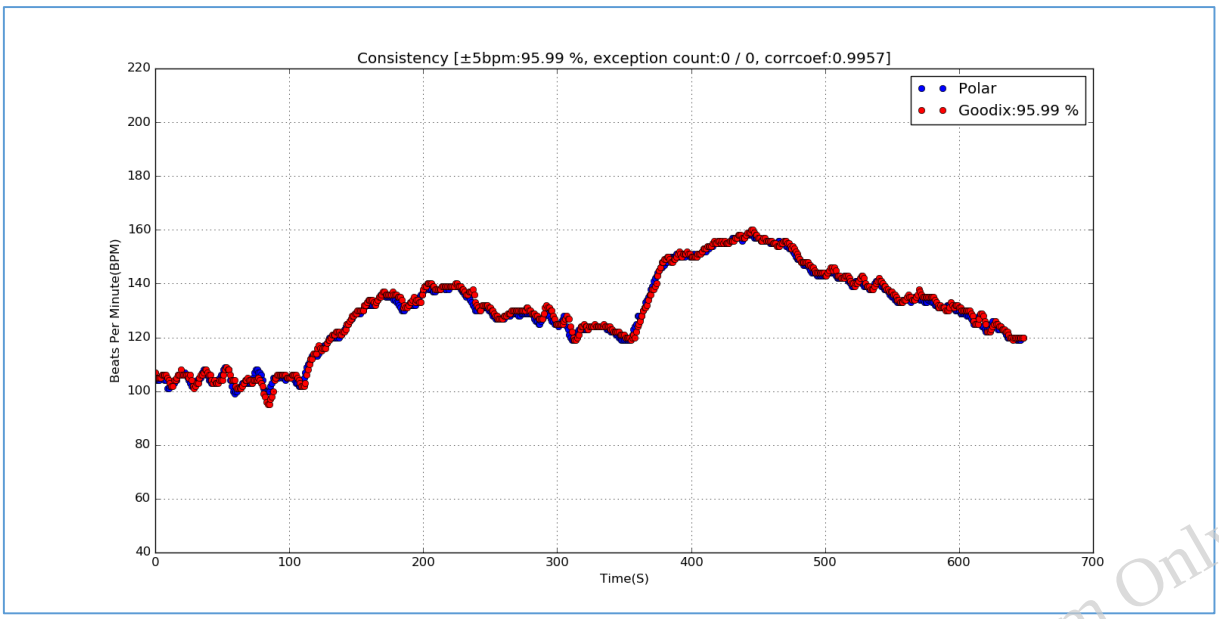

Figure 4-5 Outdoor bicycle

4.6 Wrist skipping

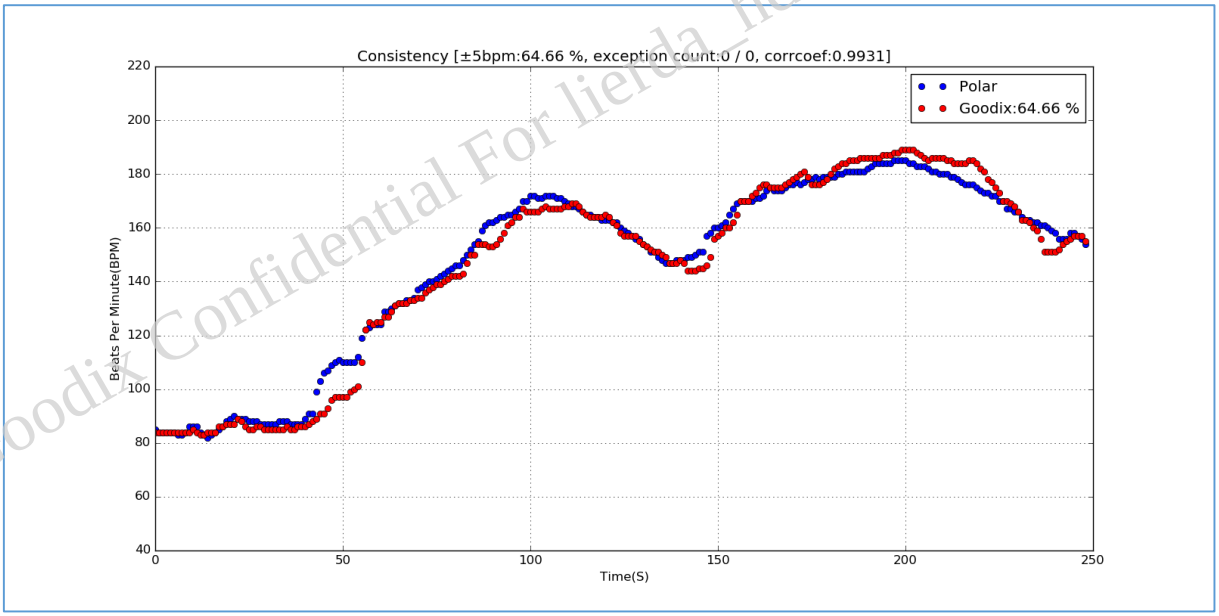

Figure 4-6 Wrist skipping

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5 Glossary and Abbreviations

Table 5-1 Glossary and Abbreviations

Name	Description
Output Time	Average output time within 1min
Deviation ratio within ±3BPM	The percentage of heart rate deviation between the test sample and the standard device within (inclusive) ±3BPM in all test data
Deviation ratio within ±5BPM	The percentage of heart rate deviation between the test sample and the standard device within (inclusive) ±5BPM in all test data
Deviation ratio within ±10BPM	The percentage of heart rate deviation between the test sample and the standard device within (inclusive) ±10BPM in all test data
Deviation ratio within ±15BPM	The percentage of heart rate deviation between the test sample and the standard device within (inclusive) ±15BPM in all test data
MAE	Mean Absolute Error, absolute error is the average value of absolute error, which is defined as, i=1, 2, 3, ...n. In a finite number of measurements, the mean absolute error is often expressed as follows: $(\sum \lvert di \rvert)/n$, where: n is the number of measurements; di is the deviation between a set of measured values and the true value)
High Confidence %	Data with high confidence level, or data that algorithm considering highly reliability.

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